9th Class • Punjab Board • PCTB New Syllabus

Chapter 4: Cell Cycle

Solved notes in book order — MCQs, short answers, detailed answers, inquisitive questions and key points.

Chapter 4 • Units 4.1 – 4.4 • Exercise solved in book order

What this chapter is about

• The cell cycle is a regular sequence of events starting after cell division to the next division: interphase (G1, S and G2) and M phase (mitosis and cytokinesis).

• Mitosis is the type of cell division in which a cell divides into two daughter cells, each with the same number of chromosomes as the parent cell. Its phases are prophase, metaphase, anaphase and telophase, followed by cytokinesis.

• Meiosis is the type of cell division in which each daughter cell receives half the number of chromosomes; it takes place in two divisions, meiosis-I and meiosis-II.

• The exercise (10 MCQs, 12 short answers, 5 detailed answers and 2 inquisitive questions) is solved here in the same order as in the book.

A. Multiple Choice Questions

Tick the correct option. Read the reason after each answer — it also tells you why the other options are wrong.

1. In which phase of cell cycle, maximum growth occurs in cell?

a) M phase
b) S phase
c) G1 phase
d) G2 phase

Answer: c) G1 phase — G1 (First Gap Phase) is also called the growth phase: the cell makes new proteins and organelles and grows in size. In G2 the cell only continues to grow and prepares for division.

2. In which phase of cell cycle, the chromosomes duplicate?

a) Mitosis
b) G1 phase
c) G2 phase
d) S phase

Answer: d) S phase — In the S (Synthesis) phase the DNA of each chromosome is replicated, so every chromosome comes to have two sister chromatids.

3. Which of the following is NOT a characteristic of mitosis?

a) It occurs in somatic cells
b) It results in genetically identical daughter cells
c) The chromosome number is halved in daughter cells
d) It results in the formation of two daughter cells

Answer: c) The chromosome number is halved in daughter cells — Halving of the chromosome number is the characteristic of meiosis. In mitosis the chromosome number of the daughter cells is the same as that of the parent cell.

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4. At which stage of mitosis chromosomes line up in the centre?

a) Prophase b) Metaphase
c) Anaphase d) Telophase

Answer: b) Metaphase — In metaphase the chromosomes arrange themselves along the equator of the cell and form the metaphase plate.

5. If you observe a cell in which nuclear membrane is reforming around two sets of chromosomes, what stage of cell cycle is this?

a) Anaphase b) Telophase
c) Prophase d) Metaphase

Answer: b) Telophase — In telophase the chromatids have reached the poles, a new nuclear envelope forms around each set of chromosomes and the nucleolus reappears.

6. How does the centrosome contribute to mitosis?

a) Initiates DNA replication b) Makes mitotic spindle
c) Forms the nuclear envelope d) Duplicates organelles

Answer: b) Makes mitotic spindle — The centrosome (a pair of centrioles) duplicates and moves to opposite poles, where it forms the spindle fibres that pull the chromosomes.

7. Centrosomes make mitotic spindle in;

a) Animal cells b) Plant cells
c) Prokaryotic cells d) All of these

Answer: a) Animal cells — Plant cells have no centrosome; their mitotic spindle is formed by the aggregation of spindle fibres present in the cytoplasm.

8. An organism has 4 pairs of chromosomes. After meiosis-I, how many chromosomes and chromatids will be present in each daughter cell?

a) 8 chromosomes and 16 chromatids b) 4 chromosomes and 8 chromatids
c) 4 chromosomes and 4 chromatids d) 8 chromosomes and 8 chromatids

Answer: b) 4 chromosomes and 8 chromatids — 4 pairs means 2n = 8 chromosomes. After meiosis-I the number becomes half, i.e. 4 chromosomes, and each chromosome still has two chromatids, i.e. 4 x 2 = 8 chromatids.

9. Which event is unique to meiosis but not mitosis?

a) DNA replication b) Chromosome alignment
c) Crossing over d) Nuclear division

Answer: c) Crossing over — Crossing over (exchange of segments between homologous chromosomes) occurs only in prophase-I of meiosis, so it is the source of new gene combinations.

10. Why is meiosis-II necessary after meiosis-I?

a) To replicate chromosomes b) To reduce chromosome number
c) To separate sister chromatids d) To ensure genetic recombination

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Answer: c) To separate sister chromatids — Meiosis-I separates the homologous chromosomes but the sister chromatids stay together; meiosis-II separates them, so that each gamete gets one chromatid of each chromosome. There is no DNA replication before meiosis-II.

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B. Write Short Answers

1. Enlist the events that occur during the G1 phase of interphase.

G1 (First Gap Phase) starts from the end of the mitosis phase. It is also called the growth phase. During this phase:

1. the cell makes new proteins and organelles and so grows in size;

2. the cell also makes the enzymes that are required in the S phase for the replication of DNA;

3. the cell remains metabolically very active and carries out its normal functions; and

4. the cell checks its own condition and, if the conditions are not suitable, it may stop dividing temporarily or permanently and enter the G0 phase.

2. What is the main purpose of the S phase in the cell cycle?

The main purpose of the S (Synthesis) phase is the replication of DNA. During this phase the DNA of each chromosome is copied, which results in the duplication of chromosomes — each chromosome then consists of two sister chromatids joined at a centromere. The total number of chromosomes in the cell, however, remains the same. In this way the S phase provides a complete copy of the genetic material for the daughter cells.

3. During which phase of mitosis sister chromatids separate?

Answer: Anaphase

During anaphase the centromeres of the chromosomes divide and the sister chromatids separate. The spindle fibres pull them towards the opposite poles, so one chromatid of each chromosome goes to each pole.

4. How does crossing over contribute to genetic variation in meiosis?

Crossing over takes place during prophase-I of meiosis. The homologous chromosomes come close together in pairs and their non-sister chromatids exchange segments of DNA at a point called chiasma. As a result, the chromosomes that go to the gametes carry new combinations of genes which were not present in either parent. This is a recombination of genetic material, so it produces variations in the offspring and makes every offspring slightly different from the others.

5. What is the role of spindle fibres in mitosis?

Spindle fibres are microtubules formed by the centrosome. Their roles are:

1. they attach to the kinetochore of each chromosome and arrange the chromosomes on the metaphase plate in metaphase;

2. they pull the sister chromatids apart and towards the opposite poles in anaphase; and

3. they help in the movement of chromosomes and, at the end, in the formation of the two daughter nuclei. Without spindle fibres the chromosomes cannot be

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distributed equally between the daughter cells.

6. How is cytokinesis in animal cell different from plant cell?

Cytokinesis in animal cell	Cytokinesis in plant cell
A cleavage furrow appears in the cell membrane between the two nuclei.	No cleavage furrow is formed, because of the rigid cell wall.
The furrow becomes deeper and divides the cytoplasm into two; the cell is pinched in two.	A cell plate is formed in the middle of the cell by the Golgi vesicles.
Cytokinesis occurs by furrowing (cleavage) of the cell membrane.	The cell plate grows outward and forms the middle lamella; new cell walls are laid down on it.
The two daughter cells remain separate and rounded.	The two daughter cells stay joined by the middle lamella and both get a cell wall.

7. What is the difference between prophase of mitosis and prophase-I of meiosis-I?

Prophase of mitosis	Prophase-I of meiosis-I
Homologous chromosomes do not form pairs; they lie separately.	Homologous chromosomes come close together and form pairs (this is called synapsis).
No crossing over takes place.	Crossing over takes place between non-sister chromatids, and chiasmata are formed.
The chromosomes are single in number (same as the parent cell), each with two chromatids.	The cell is diploid and the chromosome number is not reduced yet; tetrads are formed.
It is shorter and simpler.	It is long and is divided into sub-stages (leptotene, zygotene, pachytene, diplotene, diakinesis).

8. How does meiosis differ from mitosis in terms of chromosome number?

In **mitosis** the chromosome number of the daughter cells remains the same as that of the parent cell (2n to 2n). The chromosomes replicate once and the cell divides once, so both daughter cells get a full set of chromosomes and they are genetically identical to the parent cell.

In **meiosis** the chromosome number is reduced to half (2n to n). The chromosomes replicate only once, but the cell divides twice, so four daughter cells are formed and each of them has half the number of chromosomes of the parent cell. In humans, a

somatic cell has 46 chromosomes while a gamete has 23.

9. What are the key events of anaphase in mitosis?

1. The centromeres of the chromosomes divide, so the two sister chromatids of each chromosome separate from each other.
2. The spindle fibres contract and pull the chromatids (now called daughter chromosomes) towards the opposite poles of the cell.
3. The chromatids move with their centromeres first, and each pole receives one complete set of chromosomes.
4. At the end of anaphase the chromosomes gather near the two poles and the cell becomes ready for telophase.

10. What is the function of the centrosome during cell division?

The centrosome is a pair of centrioles present near the nuclear envelope of animal cells. Its functions are:
1. it duplicates before cell division and the two centrosomes move to opposite sides of the nucleus;
2. it forms the spindle fibres (mitotic spindle) which attach to the kinetochores of chromosomes; and
3. these fibres arrange the chromosomes on the metaphase plate and then pull the chromatids to the opposite poles, so the centrosome makes sure that the chromosomes are distributed equally between the daughter cells. (Plant cells have no centrosome; their spindle is formed from the cytoplasm.)

11. What are sister chromatids, and when do they separate in meiosis?

When the DNA is replicated in the S phase, each chromosome is made of two identical copies joined at a single centromere. These two copies are called sister chromatids. They contain the same genes and are formed so that each daughter cell may get one copy.
In meiosis the sister chromatids do not separate in meiosis-I; there the homologous chromosomes separate. The sister chromatids separate in **anaphase-II** of meiosis-II, when their centromeres divide and the spindle fibres pull them to the opposite poles. In this way each of the four daughter cells receives one chromatid of every chromosome.

12. How is mitosis related to the process of regeneration?

Regeneration is the replacement of lost or damaged body parts, and it takes place by mitosis. The cells near the damaged area divide by mitosis; the two daughter cells produced are genetically identical to the parent cell, so the new cells are exactly like the old ones and they fill up the gap. In this way mitosis repairs wounds, heals broken bones, replaces the worn-out cells of the skin and gut, and in animals such as starfish and Hydra, mitosis of the body cells regenerates a whole lost arm or even a whole new animal.

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Remember the difference

• G1 = growth phase (proteins and organelles are made). S = DNA replication (sister chromatids are formed). G2 = more growth, repair of DNA and preparation for division.

• Mitosis: one division, two daughter cells, chromosome number the same (2n). Meiosis: two divisions, four daughter cells, chromosome number halved (n).

• Cytokinesis: furrowing in animal cells, cell plate in plant cells.

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C. Write Answers in Detail

1. Describe the events that occur during the phases of mitosis.

Definition: mitosis is the type of cell division in which a cell divides into two daughter cells, each with the same number of chromosomes as the parent cell. It occurs in somatic cells and is also called equational division because the number of chromosomes is not changed. Mitosis has four phases: prophase, metaphase, anaphase and telophase; after these, cytokinesis divides the cytoplasm.

1. Prophase: the thread-like chromatin material condenses and makes thick, visible chromosomes. Each chromosome consists of two sister chromatids attached by a single centromere. The nuclear envelope and the nucleolus break down during prophase. The centrosome of the cell duplicates into two; the two centrosomes migrate to opposite sides of the nucleus and, while migrating, they make a network of microtubules called spindle fibres (the complete set is called the mitotic spindle). In plant cells there is no centrosome; their mitotic spindle is formed by the aggregation of spindle fibres present in the cytoplasm.

2. Metaphase: some spindle fibres bind with the chromosomes at the point of the centromere where special kinetochore proteins are present. Two spindle fibres, from both sides, bind with one chromosome. The chromosomes attached with spindle fibres arrange themselves along the equator of the cell; in this way a plate is formed which is called the metaphase plate.

3. Anaphase: the centromeres divide and the sister chromatids of each chromosome separate from each other. The spindle fibres contract and pull the chromatids (now called daughter chromosomes) towards the opposite poles of the cell. The chromosomes move with their centromeres first, so that each pole gets one complete set of chromosomes.

4. Telophase: the chromosomes reach the poles and the spindle fibres disappear. A new nuclear envelope forms around each set of chromosomes and the nucleolus reappears. The chromosomes start to uncoil and become chromatin again, so two daughter nuclei are formed.

5. Cytokinesis: after the karyokinesis (nuclear division) the cytoplasm divides. In animal cells a cleavage furrow appears in the cell membrane and deepens until the cell is pinched into two; in plant cells a cell plate is formed by the Golgi vesicles and new cell walls are laid on it. In this way two complete daughter cells are formed, each with the same number of chromosomes as the parent cell.

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2. Describe cytokinesis in animal and plant cells.

Definition: cytokinesis is the division of the cytoplasm of a cell after the division of its nucleus. It is the last step of cell division and produces two separate daughter cells.

Cytokinesis in animal cells: in animal cells cytokinesis begins in anaphase or telophase. A cleavage furrow (a shallow groove) appears in the middle of the cell membrane, exactly between the two nuclei. The furrow deepens because a ring of microfilaments below the membrane contracts and pulls the membrane inward.

Finally the membrane meets in the centre and the cell is pinched into two daughter cells. The daughter cells of animal cells are usually rounded and separate from each other.

Cytokinesis in plant cells: plant cells have a rigid cell wall, so they cannot be pinched; instead a structure called the cell plate is formed. Vesicles from the Golgi apparatus collect at the equator of the cell and fuse to form the cell plate. The cell plate grows from the centre towards the periphery and fuses with the existing cell wall. New cell walls are then deposited on both sides of the cell plate. The middle lamella is formed in the middle, so the two daughter cells remain joined to each other and both get a complete cell wall.

Animal cell cytokinesis	Plant cell cytokinesis
Occurs by furrowing (cleavage) of the cell membrane.	Occurs by the formation of a cell plate.
A ring of microfilaments pulls the membrane inward.	Golgi vesicles collect at the equator and fuse.
No new cell wall is formed.	A new cell wall and middle lamella are formed.
Daughter cells separate and become rounded.	Daughter cells remain joined by the middle lamella.

3. Describe the significance of mitosis.

Mitosis is the most common type of cell division in multicellular organisms. It keeps the chromosome number constant and produces cells that are genetically identical to the parent cell, so it is important in the following ways:

1. Growth: the body of a multicellular organism grows by mitosis. New cells are added by mitosis in the roots and shoots of plants and in all the growing parts of animals, so the organism increases in size.

2. Development: the zygote divides by mitosis again and again to form the embryo and then the whole body. In this way mitosis is the base of the development of every organ and organ system.

3. Cell replacement: the worn-out cells of the skin, the lining of the gut and the blood cells are replaced by mitosis. Old cells die and new cells, identical to them, take their place, so the tissues keep on working.

4. Repair and regeneration: mitosis heals wounds and repairs broken bones; in some animals it also regenerates lost parts, e.g. the arm of a starfish.

5. Asexual reproduction: many organisms reproduce asexually by mitosis, e.g. binary fission in Amoeba, budding in yeast, and vegetative propagation in plants — in this way the offspring are genetically identical to the parent.

6. Maintenance of chromosome number: by keeping the chromosome number the same, mitosis makes sure that the genetic information of a species is not changed from cell to cell, which is why it is called equational division.

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Note: if mitosis goes out of control and cells keep on dividing without any need, an uncontrolled mass of cells (a tumour) is formed, which may lead to cancer.

4. Describe the events that occur during the phases of meiosis-I.

Definition: meiosis is the type of cell division in which each daughter cell receives half the chromosome number of the parent cell. It was first reported in 1876 by the German biologist Oscar Hertwig. It consists of two divisions: meiosis-I and meiosis-II. Meiosis-I is the reductional division, because the chromosome number is reduced to half.

Prophase-I: it is the longest phase and is divided into five sub-stages:

(i) Leptotene: the chromatin condenses and the chromosomes become visible as long thin threads; the centrioles start moving to the poles.

(ii) Zygotene: the homologous chromosomes come close together and pair up; this pairing is called synapsis. Each pair is called a bivalent or tetrad, because it has four chromatids.

(iii) Pachytene: the chromosomes become thicker and shorter; crossing over takes place between the non-sister chromatids of the homologous chromosomes.

(iv) Diplotene: the homologous chromosomes start to separate but remain joined at the points of crossing over. These points are called chiasmata.

(v) Diakinesis: the chromosomes shorten further, the chiasmata move to the ends of the chromosomes and the nuclear envelope and nucleolus break down. The spindle fibres are formed.

Metaphase-I: the homologous chromosome pairs arrange themselves on the equator of the cell and form the metaphase plate. The two chromosomes of each pair lie on the two sides of the plate, and each is attached to the spindle fibres of its own pole.

Anaphase-I: the homologous chromosomes of each pair separate from each other and, pulled by the spindle fibres, move towards the opposite poles. Here the centromeres do not divide, so each chromosome still has two chromatids. As a result of this separation the chromosome number becomes half at each pole.

Telophase-I: the chromosomes reach the poles and the spindle fibres disappear. A new nuclear envelope is formed around each group of chromosomes and the nucleolus reappears. Then cytokinesis takes place and two daughter cells are formed, each with half the chromosome number of the parent cell. These cells then enter meiosis-II.

5. Describe the significance of meiosis.

Meiosis is the special type of cell division which produces gametes in animals and spores in plants. Its significance is:

1. Maintenance of chromosome number of the species: in animals, the gametes (sperm and egg) have half the chromosome number of the body cells. When they fuse at fertilization, the zygote gets back the full (diploid) number. So the chromosome number of a species remains the same from generation to generation.

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2. Produces genetic variation: crossing over in prophase-I and the independent assortment of homologous chromosomes in anaphase-I produce new combinations of genes. Because of this, the offspring inherit a mixture of characters from both parents and every one of them is slightly different from the others.

3. Basis of sexual reproduction: meiosis makes the haploid gametes without which sexual reproduction cannot take place. In plants, meiosis produces the spores which grow into new plants.

4. Source of new varieties and evolution: the variations produced by meiosis are the raw material of evolution. Useful combinations are selected by nature and, in agriculture, they are used to produce new varieties of crops and animals.

5. Prevents doubling of chromosomes: if gametes were formed by mitosis, the chromosome number would double at every generation and the species would be destroyed. Meiosis prevents this by halving the number before fertilization.

6. Repairs genetic mistakes: the reshuffling of genes in meiosis hides many harmful recessive genes, because a healthy copy of the gene from the other parent usually covers the faulty one.

Errors: sometimes the process of meiosis does not work correctly and the chromosomes fail to separate properly. This is called non-disjunction; as a result, a daughter cell may receive an extra chromosome or may be missing one. For example, an extra copy of chromosome 21 causes Down syndrome.

D. Inquisitive Questions

1. What role might mistakes in the cell cycle checkpoints play in the emergence of cancer?

The cell cycle has checkpoints — special control points in G1, at the end of G2 and in metaphase — where the cell checks whether its DNA is copied correctly, whether its chromosomes are attached to the spindle, and whether the conditions are suitable for division. Only when everything is correct does the cell go on to the next phase. If these checkpoints make a mistake, the following can happen:

1. Damaged DNA goes unnoticed: a cell with a damaged or wrongly copied DNA continues to divide, so the mistake is copied into all the daughter cells.

2. Cell divides without the right signals: the cell keeps on dividing even when the body does not need new cells, because the brakes of the cycle have failed.

3. Chromosomes are distributed wrongly: if the metaphase checkpoint fails, the chromatids may not be attached correctly and the daughter cells get extra or missing chromosomes.

4. Errors in genes that control the cycle are collected: the genes that control the checkpoints (such as the genes for repair proteins and for proteins that stop the cycle) are themselves changed; after several such changes the cell loses all control.

5. Tumour formation: the uncontrolled division of cells without differentiation produces a mass of cells, which is called a tumour. Some of these cells then spread to other parts of the body and the disease is called cancer. This is why a mistake in one checkpoint is the first step towards cancer.

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2. Why do skin cells divide continuously throughout an organism's existence, but nerve and muscle cells permanently exit the cell cycle?

Skin cells: the epidermis is in direct contact with the environment, so its cells are continuously rubbed off, dried and injured. They must therefore be replaced again and again: the stem cells in the deeper layer keep dividing by mitosis, and the new cells move outward and become the protective layer. So continuous division is a need of the skin's function.

Nerve and muscle cells: these cells are formed during development and then permanently leave the cell cycle (they enter the G0 phase). Their reasons are:

1. They are highly specialized: once a neuron has made its long axon and its connections, or a muscle cell has arranged its contractile proteins, it cannot divide without disturbing its structure and function.

2. Their connections must not change: the network of neurons in the brain and the arrangement of muscle fibres in the heart are fixed; if these cells divided in an uncontrolled way, the network and the coordinated action would be lost.

3. They are long-lived: many of these cells live as long as the organism does, so there is no need for constant replacement, and their energy is used for their work rather than for division.

4. They can be repaired but not replaced: in peripheral nerves the axon can regrow, and a limited number of new neurons is produced in some parts of the brain, but in general a lost neuron or a dead cardiac muscle cell is replaced by scar tissue and not by division.

Conclusion: the difference is related to the function and life-span of the tissue. Cells whose work needs constant replacement (skin, gut lining, blood) keep dividing, while highly specialized, long-lived cells (neurons and muscle cells) stay in G0 and do not divide again.

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Points to Remember

Cell cycle is a series of events starting after cell division to the next division. It has two main phases: interphase and M phase (mitosis and cytokinesis).

Interphase takes about 90% of the total cell cycle time and has three phases: G1 (growth phase — proteins and organelles are made, cell grows in size), S (synthesis phase — DNA is replicated and chromosomes are duplicated) and G2 (more growth, repair of DNA and preparation for division). Cells that stop dividing enter the G0 phase.

Mitosis is the type of cell division in which a cell divides into two daughter cells, each with the same number of chromosomes as the parent cell. It occurs in somatic cells.

Phases of mitosis: prophase (chromatin condenses into chromosomes, nuclear envelope and nucleolus break down, spindle fibres are formed), metaphase (chromosomes line up on the metaphase plate), anaphase (centromeres divide and chromatids move to the poles) and telophase (nuclear membranes re-form around the two sets of chromosomes, nucleolus reappears).

Cytokinesis is the division of the cytoplasm: by furrowing in animal cells and by a cell plate in plant cells. It occurs at the end of the division and two new cells are formed.

Significance of mitosis: growth, development, replacement of worn-out cells, repair and regeneration, and asexual reproduction; it keeps the chromosome number constant.

Meiosis is the type of cell division in which each daughter cell receives half the number of chromosomes of the parent cell; it was first reported by Oscar Hertwig in 1876. It has two divisions, meiosis-I (reductional) and meiosis-II (equational).

Prophase-I has five sub-stages — leptotene, zygotene (synapsis), pachytene (crossing over), diplotene (chiasmata) and diakinesis. In anaphase-I the homologous chromosomes separate, so the chromosome number becomes half.

Sister chromatids separate in anaphase-II of meiosis-II (in mitosis they separate in anaphase).

Significance of meiosis: it maintains the chromosome number of the species, produces genetic variation by crossing over and independent assortment, makes sexual reproduction possible, gives new varieties for evolution and agriculture, and prevents the doubling of chromosomes.

Mitosis	Meiosis
A parent cell divides only once; two daughter cells are produced.	A parent cell undergoes two divisions; four daughter cells are produced.
Chromosome number in daughter cells remains the same as the parent cell.	The chromosome number is reduced by half in daughter cells.
Variations are not generated.	Variations occur due to crossing over.

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Mitosis	Meiosis
Occurs in somatic cells.	Occurs in germ line cells.
Homologous chromosomes do not form pairs.	Homologous chromosomes form pairs.
No crossing over occurs during prophase.	Crossing over occurs during prophase-I.
Single chromosome aligns to form a metaphase plate.	Homologous pairs align to form a metaphase plate.
During anaphase, chromosomes break and individual chromatids are pulled towards poles.	During anaphase-I, individual chromosomes are pulled towards poles.
Occurs for growth, development and maintenance of multicellular organisms.	Occurs for producing gametes in animals and spores in plants for sexual reproduction.

Table: Difference between mitosis and meiosis (textbook table)

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d).

• Short question: 2 to 4 sentences; write the name of the phase first and then its events.

• Difference question: always draw a two-column table with headings.

• Detailed answer: write the definition, then the phases or points in order, and end with significance. Diagrams of mitosis and meiosis carry extra marks.